Data file formats and Jupyter

Units 17–19 · syllabus 2024 · module marks 8, doubled to 16

HOW THIS PAPER WORKS

The syllabus gives this module 8 marks. This paper gives **twice** that (16 marks). Every file format named in unit 17 (csv, xls, tab, dat) is tested.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: MAT4CJ205 Python and SageMath (Apr 2026); CSC1MN106 Computer Fundamentals with MS Excel, SPSS (Nov 2024).

Time 1 hour

Maximum 16 marks

Section A: answer all six. Section B: answer one of two. Assume the files are in the same folder as the notebook.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 12.

1. What is the difference between a Series and a DataFrame in pandas? Give one example of each.

PYQ A26 B16 · adapted · U17

2. A capacitor experiment is saved as a comma-separated file `cap.csv`. Write the one line of code that loads it into a DataFrame `df`.

   PYQ A26 Q6 · adapted · U17

3. A phone sensor file `accel.tab` is separated by tab characters. Write the code that loads it into `df`, and explain the argument you must pass to `pd.read_csv` or `pd.read_table` to make the tabs work.

   Instructor · U17

4. Name the function used to read an Excel `.xls` or `.xlsx` file. What must be installed on the machine first if the file is an `.xls` from an older Excel version?

   PYQ CSC1MN106 N24 B12 · adapted · U17

5. A plain-text data file `spring.dat` has values separated by spaces. Explain why `sep=' '` or `sep=r'\s+'` is used when reading it, and state what happens if the default comma separator is kept.

   Instructor · U17

6. Given `df` with a row index of 0, 1, 2, 3, write the code for (a) the value in row 2, column `'Period'` using `loc`, and (b) the first two rows and the first column using `iloc`. Say in one line how `loc` and `iloc` differ.

   Instructor · U18

Section B · Descriptive

Answer one of the two. Carries 4 marks. Total 4.

7. Explain the work environments: what Jupyter Notebook is, how Anaconda provides it, and how Google Colab differs from a local install. Name one reason a physics student would prefer Colab.

Instructor · U19

8. A student has three files for the same experiment: `run1.csv`, `run2.dat` and `run3.tab`, with columns `length` and `period`. Write a loop that reads all three into one list of DataFrames, then uses `pd.concat` to join them, and explain why the separators are different in each file.

Instructor · U17, U18

COVERAGE

UNIT	TOPIC	QUESTIONS
17	Series and DataFrames, csv, xls, tab, dat formats	1, 2, 3, 4, 5, 8
18	loc and iloc, operations on DataFrames	6, 8
19	Jupyter Notebook, Anaconda, Google Colab	7

Total marks: 12 (Section A) + 4 (Section B) = 16.